

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

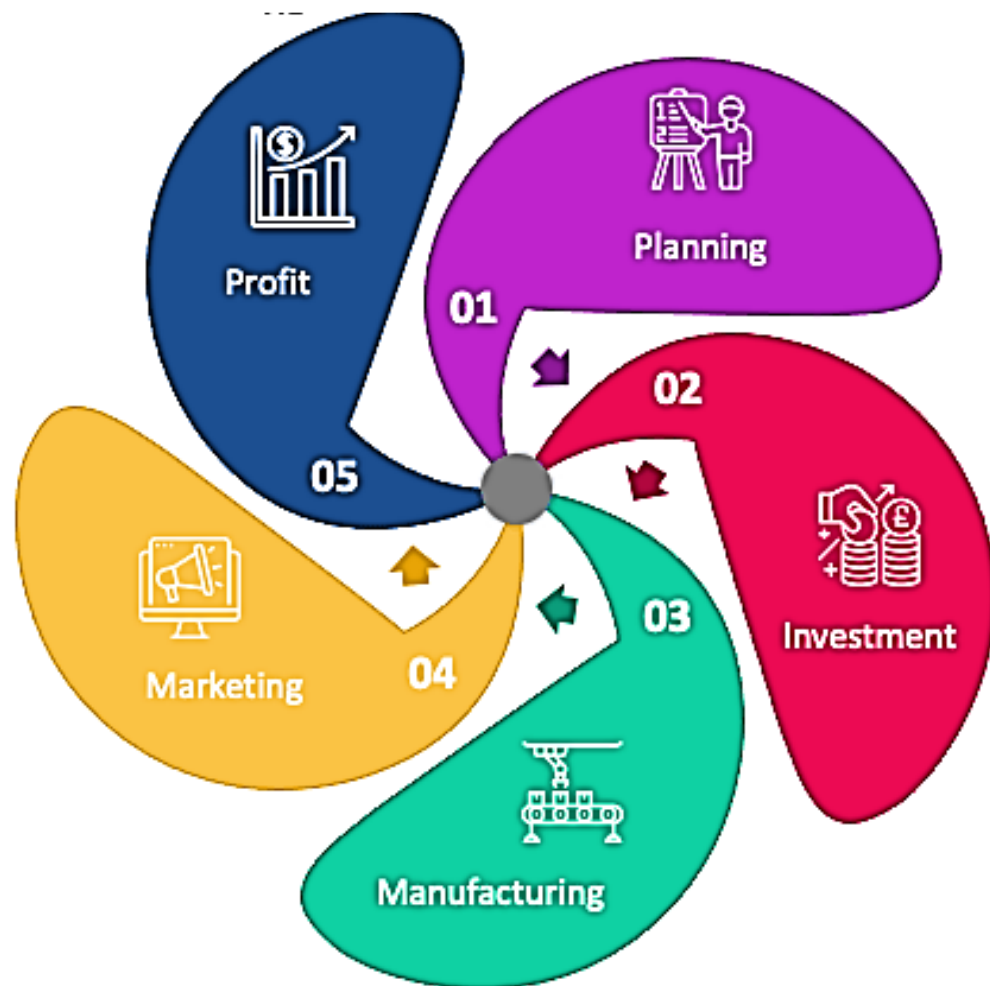
In the name of Allah, Most Gracious,
Most Merciful.



Course HS - 402, Engineering Economics, CR 1, (2+0)/week



Engineering Economics (HS - 407)



Course Instructor

Engr. Yasir Hamid

Lecturer

**Department of Mechanical Engg.
HITEC University, Taxila Cantt**

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Course Outline & Reference Books

Join Google Classroom:

<https://classroom.google.com/c/NzYwNDU5NjkwNDg2?cjc=rk4peiq>

rk4peiq



Course Outline & Reference Books

Reference Books

- 1. Engineering Economy by William G. Sullivan, Elin W. Wicks, C. Patrick Koelling**
- 2. Economics by Campbell R Cornell, Stanley L. Bruce**
- 3. Project Management, the managerial process by Erik W. Larson, Clifford F. Gray**
- 4. Karl E. Case's Principles of Economics (10th Edition)**

Office Visiting Hours

- ❖ **Office 08 MED,**
- ❖ **Phone: Ext 327/329**
- ❖ **Office Hours: 0830 - 1530 hrs.**
- ❖ **Lunch Break 1230 - 1330 hrs.**
- ❖ **Student Hours: 0930 - 1630 hrs. Tuesday**

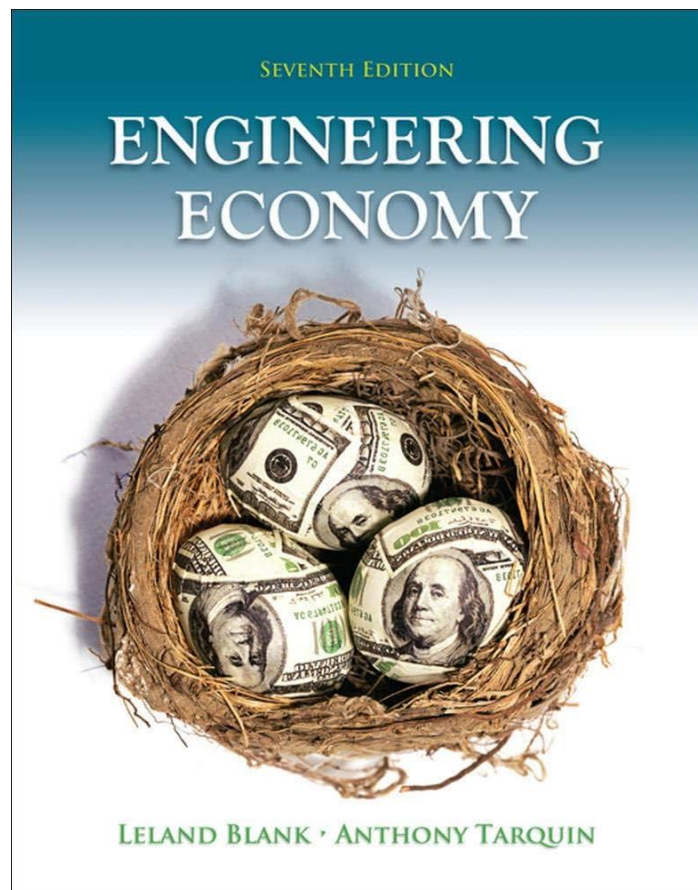


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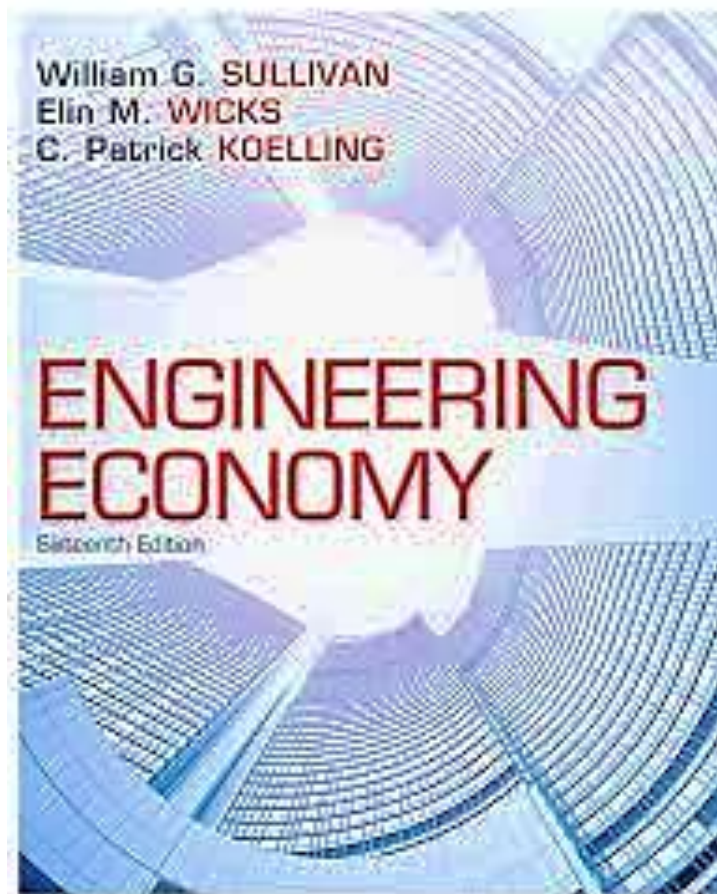


Course Outline & Reference Books

Reference Book 1



Reference Book 2



Reference Book 3





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Course Outline & Grading Policy

Assignments=10% Quizzes=05% for 1/CH → 2 Quizzes, Project=10% Mid Exam=30% each, Finals=45%

Quiz 1 **Week 03**

Assignment 1 **Week 04**

Quiz 2 **Week 05**

Assignment 2 **Week 06**

Mid Exam **Week 08**

Quiz 3 **Week 10**

Assignment 3 **Week 11**

Quiz 4 **Week 12**

Assignment 4 **Week 13**

Project **Week 14**

Presentation **Week 15**

Finals **Week 17**



Course Outline & Class Policy

- ❖ You are highly encouraged to ask questions and discuss the course material with me and your classmates.
- ❖ No cheating, fabrication, falsification, forgery and computer misuse.
- ❖ Please switch off/put on silent your cell phones.
- ❖ Adhere to the given date for assignment submission.
- ❖ Academic Honesty: Plagiarism will not be tolerated at any level.
- ❖ Extra Help: Do not hesitate to come to my office during office hours or by appointment to discuss a problem or any aspect of the course.
- ❖ University Attendance Policy: Students are expected to attend classes regularly. Minimum attendance must be 75%.



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Course Outline & Important Note

Make-Up Of Any Exams / Quizzes / Sessional Will Not Be Entertained – Make Sure To Attend It As Scheduled

Make Sure To Comply With The Attendance Policy (75% minimum). No Relaxation Will Be Granted

**Make Habit of Self-Study – Make Habit of Reading Books
Make Use of the Power of Internet for Research**

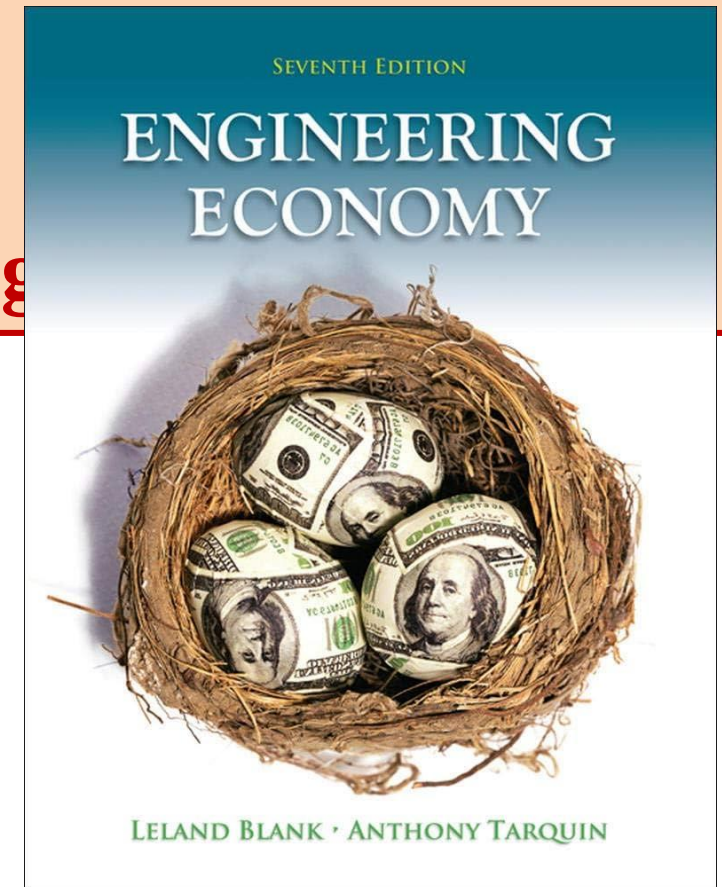


Introduction to Engineering Economics

Chapter # (1)

The Scope and Nature of Engineering Economics

The Foundation of Engineering Economics by Leland Blank





Introduction to Engineering Economics (Economic and Market Forces)

1. **Economic forces** are mechanisms that ration scarce goods OR Some of the most notable economic forces include inflation, interest rates, government spending, tax rates, and the unemployment rate.
2. **Market force** is an economic force that is given relatively free rein by society to work through the market OR The definition of market forces is that it is the economic factors that influence the price and quantity of goods and services in a market. Examples of market forces are competition, consumer preferences, technological advancements, economic growth, and government regulations
3. The **invisible hand** is the price mechanism that guides our actions in a market. The invisible hand is an example of a market force.

The invisible hand theory, a concept popularized by Adam Smith, suggests that a free market, where individuals pursue their own self-interest, can lead to beneficial outcomes for society as a whole, without the need for central planning or intervention

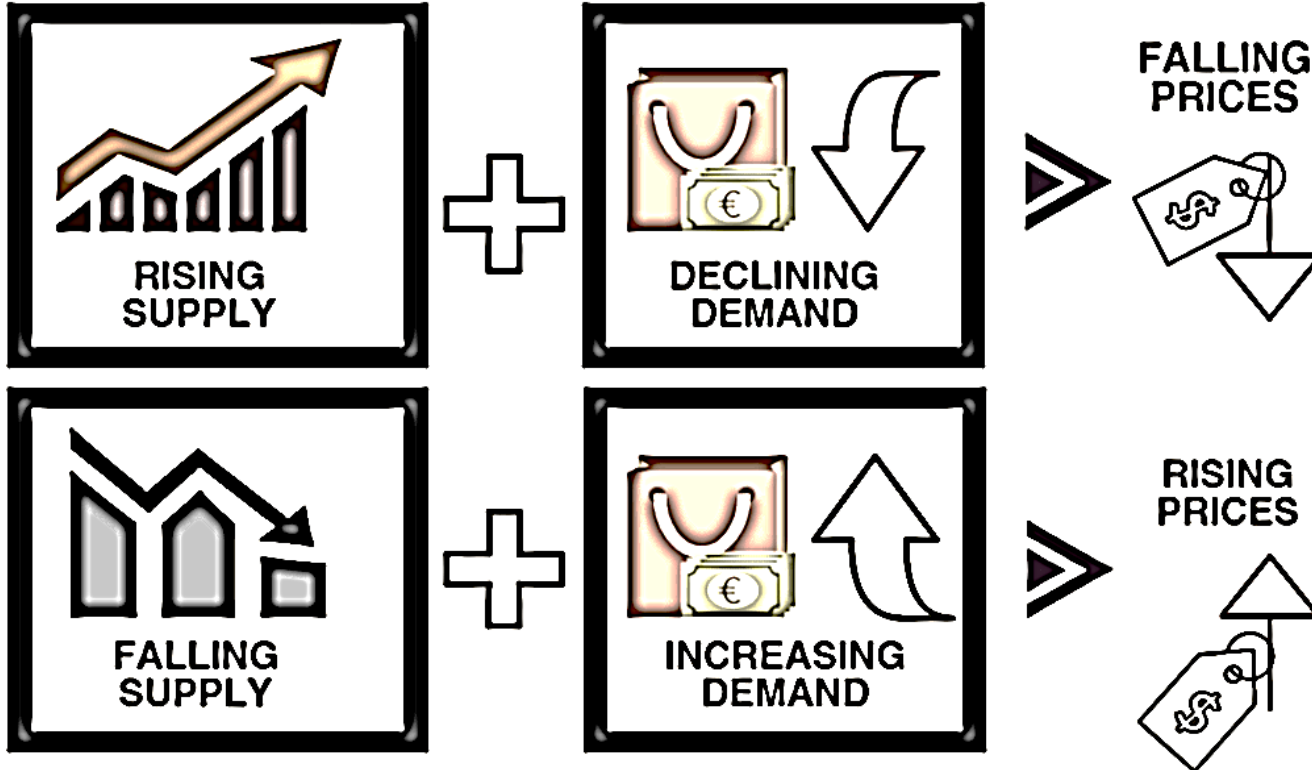
What happens in society can be seen as a reaction to, and interaction of:

- ✓ Economic forces
- ✓ Social forces
- ✓ Historical forces



Introduction to Engineering Economics (Economic and Market Forces)

MARKET FORCES



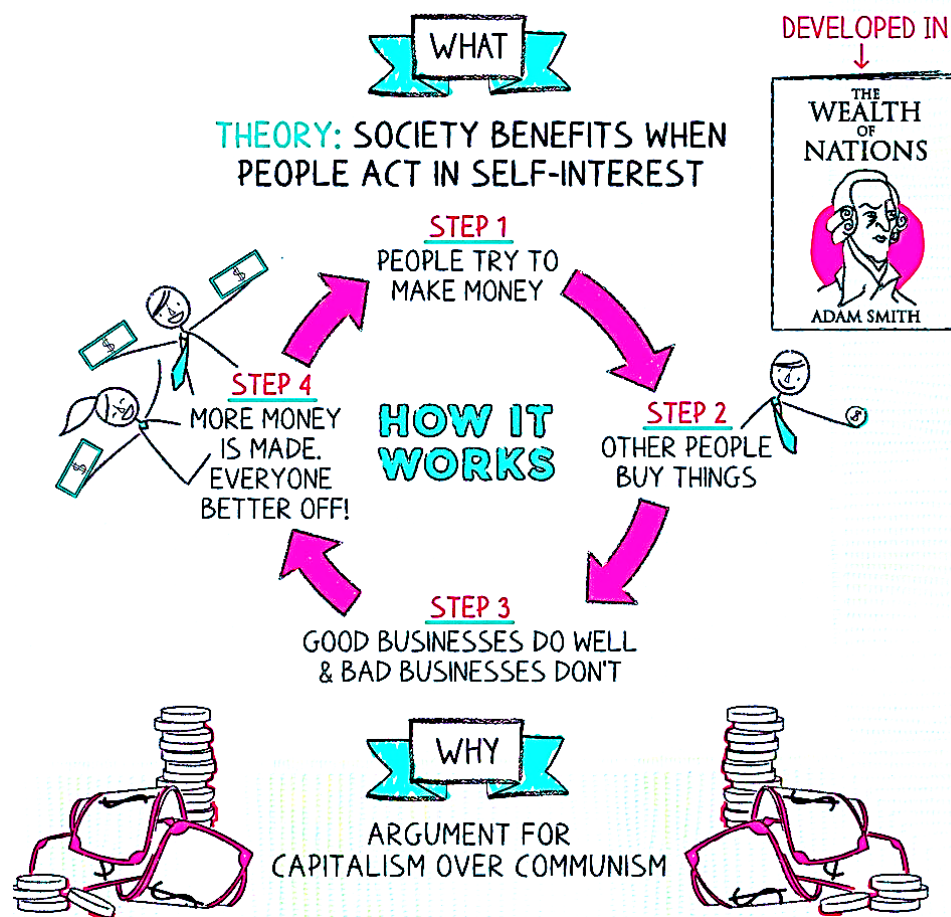
✓ If there is a shortage, **prices rise**

✓ If there is a surplus, **prices fall**

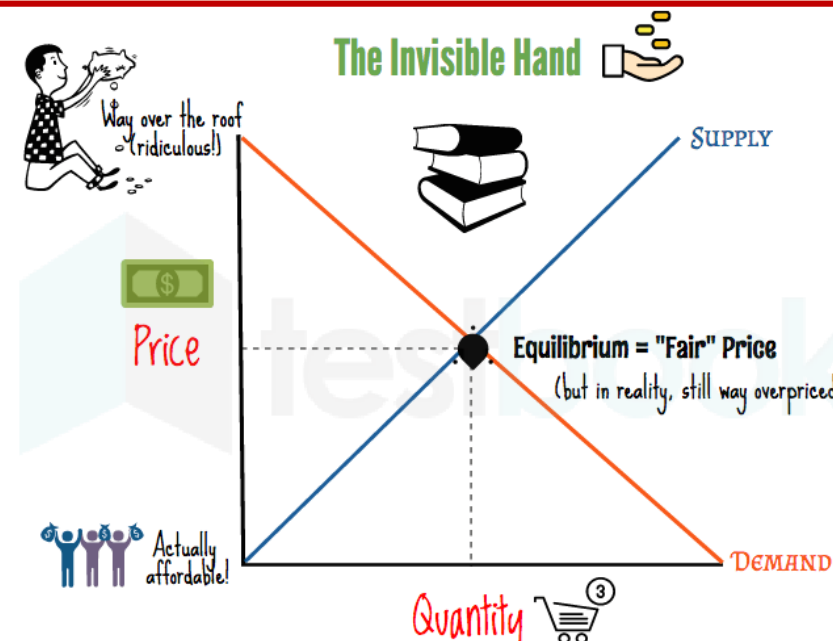


Introduction to Engineering Economics (Economic and Market Forces)

INVI\$IBLE HAND



It is not from the benevolence of the butcher, brewer, or the baker, that we expect our dinner, but from their regard to their own interest.

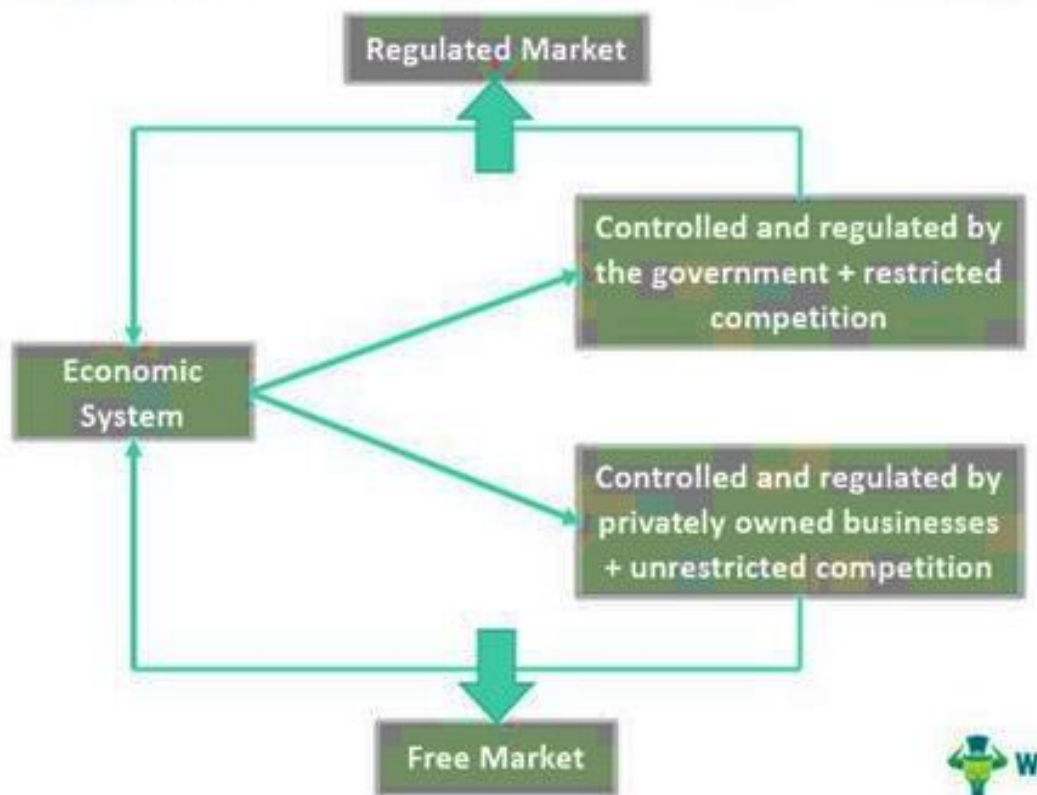


Social, cultural, and political forces influence market forces
Political and social forces often work together against the invisible hand



Introduction to Engineering Economics (Economic and Market Forces)

Regulated Market vs Free Market



Free Market Economy Pros and Cons

Economic freedom and choice

Incentives for businesses to be efficient and innovative

Greater competition can lead to lower prices for consumers

PROS

CONS

Limited government regulation can lead to monopolies or other abusive business practices

Lack of government intervention can lead to large income disparities

Environmental protection may be weaker in a free market economy





Introduction to Engineering Economics (Economic and Market Forces)

Feature	Fascism	Capitalism	Socialism	Communism
Political System	Authoritarian, far-right	Democratic or varies	Often democratic (in modern forms)	Authoritarian (in practice)
Economic System	State-directed capitalism	Free market capitalism	Mixed economy	State/collective ownership
Focus	Nationalism, order, power	Individual freedom, profit	Social welfare, fairness	Classless society, equality
Role of State	Total control (especially political)	Minimal (in theory)	Active (taxes, regulations)	Total (economy and society)
Rights & Freedoms	Suppressed	Protected	Protected but regulated	Suppressed (in practice)

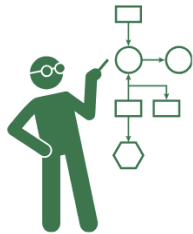


Introduction to Engineering Economics (Economic Insights)

- Theories **tie together** economist's **terminology and knowledge** about **economic institutions**.
 - Theories are **too abstract to apply in specific cases** and are often **embodied in economic models and principles**
 - An **economic model is a framework** that places the **generalized insights of the theory** in a more **specific contextual setting**.
 - An **economic principle** is a commonly **held insight** stated as a **law or general assumption**.
- According to the invisible hand theory, a market economy, through the price mechanism, will allocate resources efficiently
 - Prices *fall* when quantity supplied is greater than quantity demanded
 - Prices *rise* when the quantity demanded is greater than the quantity supplied
 - Efficiency means achieving a goal as cheaply as possible



Introduction to Engineering Economics (Economic Insights) **Economic Theory**



**Structured
Framework**



**Analyzes Economic
System**



**Helps in Better
Decision-Making**

Theory, Principles, and Models



An economic theory evolves from a hypothesis that accumulates favorable results after continued testing against the facts.

Economic laws and principles are widely accepted theories that have been well tested and widely accepted.

An economic model is a simplified representation of how something works using a combination of laws or principles.

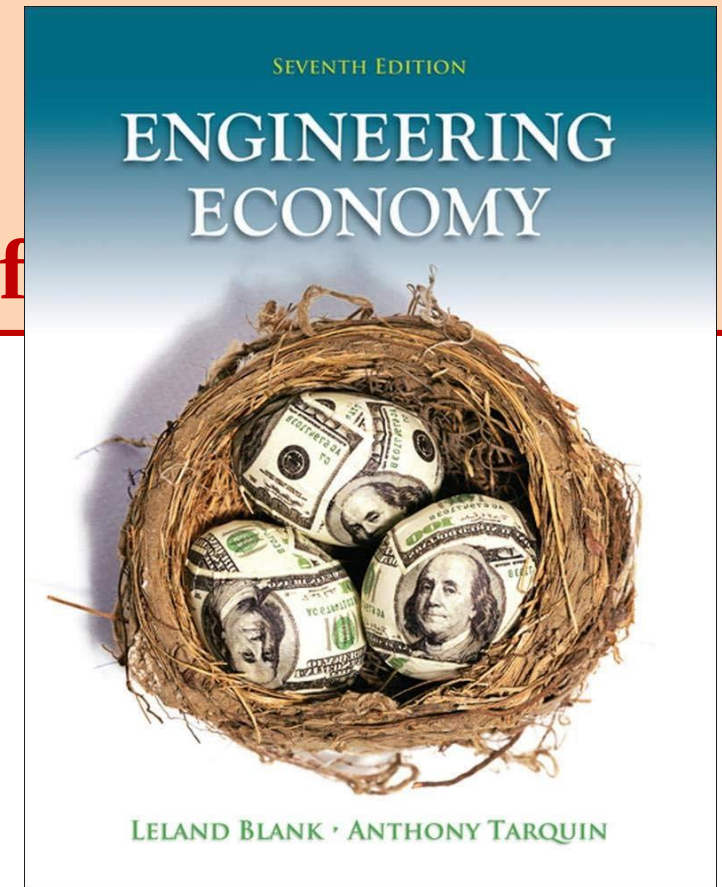


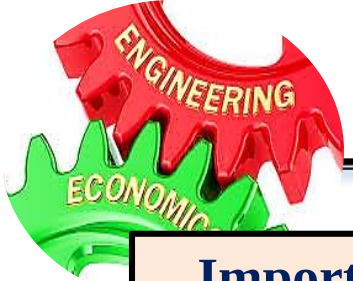
Time Value of Money & Interest Calculations

Chapter # (2)

Factors: How Time and Interest Aff

The Foundation of Engineering Economics by Leland Blank





Important Terminologies

- ✓ Adjusted (cost) basis
- ✓ **Unadjusted cost basis**
- ✓ Book Value (BV)
- ✓ **Market Value (MV)**
- ✓ -----
- ✓ Recovery Period
- ✓ **Recovery rate**
- ✓ Salvage Value
- ✓ **Useful life**
- ✓ -----
- ✓ Interest
- ✓ **Interest Rate**
- ✓ Simple Interest
- ✓ **Compounds Interest**

Unadjusted (cost) basis

The initial value of acquiring an asset (purchase price + any sales tax) including transportation expenses and other normal costs of making the asset serviceable for its intended use; also called unadjusted cost basis.

OR

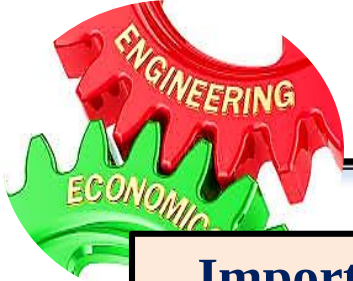
The original, raw cost of something at the time it was incurred. No modifications are made for inflation, location, or market changes. Also known as: Nominal cost or historical cost.

Adjusted (cost) basis

The original cost basis of the assets, adjusted by allowable increase or decrease is used to compute depreciation deductions or additions

OR

The cost modified to reflect current conditions like inflation, currency conversion, taxes, depreciation, or other relevant economic factors.



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Book Value

Book value is the value of an asset or a company according to its financial statements (books).

$$\text{Book Value} = \text{Total Assets} - \text{Total Liabilities}$$

The original cost basis of the property, including any adjustments, less all allowable depreciation deductions.

Market Value

Market value is the current price at which an asset or a company can be bought or sold in the open market

$$\text{Market Value} = \text{Stock Price} \times \text{Number of Outstanding Shares}$$

This is also known as market capitalization for a company.
The present value of what will be received through ownership of the property



Time Value of Money and Interest Rate (Book Value)

It's what shareholders would theoretically receive if the **company were liquidated** (all assets sold and all debts paid).

BOOK VALUE Share Capital + Retained Earnings	=	Total Assets Current Assets + Non-Current Assets	-	Total Liabilities Current Liabilities + Non-Current Liabilities
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A company owns \$5 million in assets and owes \$2 million.
Book value = \$5M – \$2M = \$3 million

A company owns machinery purchased for \$100,000. After 5 years, it's depreciated to \$60,000.

- Book value of the machinery = \$60,000, even if it could sell for more or less today.



Time Value of Money and Interest Rate (Market Value)

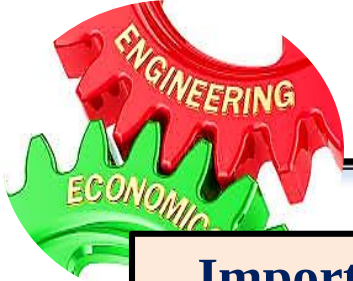
If a company has 1 million shares, and each is trading at \$5:

$$\text{Market value} = 1\text{M} \times \$5 = \$5 \text{ million}$$

1 million shares outstanding

Stock price = \$20

$$\text{Market Value} = 1,000,000 \times 20 = \$20 \text{ million}$$



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Recovery Period

The number of years over which the basis of the property is recovered through accounting process.

Recovery Rate

A percentage (expressed in decimal form) for each year of the recovery period which is applied for the deduction of depreciation.

The percentage of the initial value that is recovered after a loss.

If you lost \$500 on an investment and later recovered \$400 of that:

$$\text{Recovery Rate} = (\text{Amount Recovered} / \text{Amount Lost}) \times 100$$

$$\text{Recovery Rate} = (400 / 500) \times 100 = 80\%$$

So, the recovery rate is 80%—meaning you recovered 80% of the loss.



Recovery Period

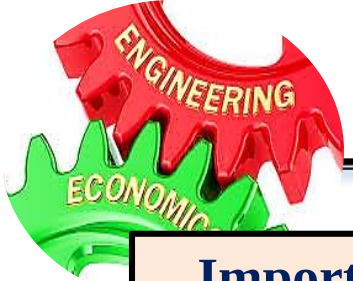
The time it takes to recover an investment or get back to a profitable state after a loss or damage.

If you lose \$1,000 on an investment and it takes you 2 years to make that \$1,000 back:

👉 The recovery period is 2 years.

If a factory burns down, and it takes 6 months to rebuild and get it back to full production:

👉 The recovery period is 6 months.



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Salvage Value

The estimated value of a property at the end of its useful life. It is the expected selling price of a property when the asset can no longer be used.

You buy a machine for \$10,000, and after using it for several years, you think you can sell it for \$1,000 when it's no longer useful, That \$1,000 is the salvage value of the machine.

Useful Life

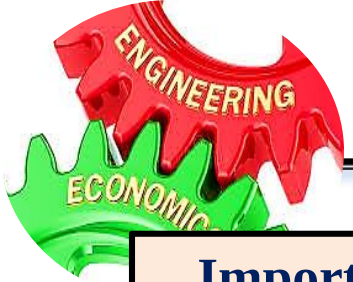
The expected (estimated) period that a property will be used in a trade or business to produce income. It is not how long the property can last but how long it can be used productively.

When a company buys an asset, it can depreciate it over time based on its useful life and the salvage value.

$$\text{Depreciation per Year} = (\text{Cost of Asset} - \text{Salvage Value}) / \text{Useful Life}$$

if the machine cost \$10,000, has a salvage value of \$1,000, and a useful life of 5 years

$$\text{Depreciation per Year} = (\$10,000 - \$1,000) / 5 = \$1,800 \text{ per year}$$



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Interest

It is the fee which is charged in using someone else's money

If you borrow \$1,000 from a bank for a year at an interest rate of 5%, the interest you will pay is:

$$\text{Interest} = \text{Principal} \times \text{Interest Rate}$$

$$\text{Interest} = \$1,000 \times 5\% = \$50$$

So, you would pay \$50 as interest over the year in addition to the \$1,000 you borrowed.

Interest Rate

The percentage of money which is charged as Interest is called Interest Rate

If you borrow money with an interest rate of 5% per year, that means you pay 5% of the principal as interest each year.



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Simple Interest

Fixed Percentage of the Principal (amount of money borrowed) multiplied by the life of the loan

Simple Interest (SI) = Principal Amount × Interest Rate × Number of Time Periods

If you invest \$1,000 at an interest rate of 5% per year for 3 years, the interest would be:

$$\text{SI} = \$1,000 \times 5\% \times 3 = \$150$$

Compound Interest

Interest that is calculated on both the principal and the interest that has accumulated from previous periods.

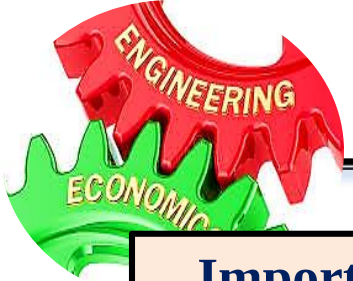
$$\text{Compound Interest (CI)} = \text{Principal Amount} \times \left(1 + \frac{\text{Interest Rate}}{n}\right)^{n \times \text{Time}} - \text{Principal Amount}$$

Principal = The initial amount of money

Rate = The interest rate per period

n = The number of times interest is compounded per year (e.g., annually, quarterly)

Time = The number of years



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Compound Interest

Interest that is calculated on both the principal and the interest that has accumulated from previous periods.

$$\text{Compound Interest (CI)} = \text{Principal Amount} \times \left(1 + \frac{\text{Interest Rate}}{n}\right)^{n \times \text{Time}} - \text{Principal Amount}$$

Let's say you invest \$1,000 at an interest rate of 5% per year, compounded annually for 3 years.

$$CI = \$1,000 \times (1 + 5\%)^3 - \$1,000$$

$$CI = \$1,000 \times (1.05)^3 - \$1,000$$

$$CI = \$1,000 \times 1.157625 - \$1,000$$

$$CI = \$1,157.63 - \$1,000 = \$157.63$$

So, after 3 years, you would earn \$157.63 in interest with compound interest, which is slightly more than the \$150 from simple interest.

$$\text{Total amount after 3 years} = \$1,000 + \$157.63 = \$1,157.63$$